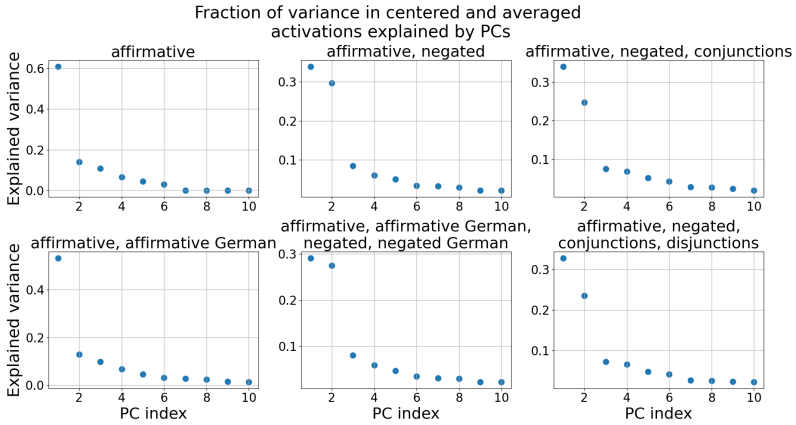


Figure 4: The fraction of variance in the centered and averaged activations $\tilde{\boldsymbol{\mu}}_i^+$, $\tilde{\boldsymbol{\mu}}_i^-$ explained by the Principal Components (PCs). Only the first 10 PCs are shown.

with the truth by averaging the activations:

$$\tilde{\boldsymbol{\mu}}_i^+ = \frac{2}{n_i} \sum_{j=1}^{n_i/2} \tilde{\mathbf{a}}_{ij}^+ \qquad \tilde{\boldsymbol{\mu}}_i^- = \frac{2}{n_i} \sum_{j=1}^{n_i/2} \tilde{\mathbf{a}}_{ij}^-, \tag{6}$$

where $\tilde{\mathbf{a}}_{ij}^+$ and $\tilde{\mathbf{a}}_{ij}^-$ are the centered activations corresponding to true and false statements, respectively. We then perform PCA on these preprocessed activations, including different statement types in the different plots. For each statement type, there are six topics and thus twelve centered and averaged activations $\tilde{\boldsymbol{\mu}}_i^\pm$ used for PCA.

Figure 4 illustrates our findings. When applying PCA to affirmative statements only (top left), the first PC explains approximately 60% of the variance in the centered and averaged activations, with subsequent PCs contributing significantly less, indicative of a one-dimensional affirmative truth direction. Including both affirmative and negated statements (top center) reveals a two-dimensional truth subspace, where the first two PCs account for more than 60% of the variance in the preprocessed activations. Note that in the raw, non-preprocessed activations they account only for $\approx 10\%$ of the variance. We verified that these two PCs indeed approximately correspond to $\mathbf{t}_G$ and $\mathbf{t}_P$ by computing the cosine similarities between the first PC and $\mathbf{t}_G$ and between the second PC and $\mathbf{t}_P$, measuring cosine similarities of 0.98 and 0.97, respectively. As shown in the other panels of Figure 4, adding logical conjunctions, disjunctions and statements translated to German does not increase the number of significant PCs beyond two, indicating that two principal components sufficiently capture the truth-related variance, suggesting only two truth dimensions.

4.2 Generalization of different truth directions

To further investigate the dimensionality of the truth subspace, we examine two aspects: (1) How well different truth directions $\mathbf{t}$ trained on progressively more statement types generalize; (2) Whether the activations of true and false statements remain linearly separable along some direction $\mathbf{t}$ after projecting out the 2D subspace spanned by $\mathbf{t}_G$ and $\mathbf{t}_P$ from the training activations. Figure 5 illustrates these aspects in the left and right panels, respectively. We compute each $\mathbf{t}$ using the supervised learning approach from Section 3, with all polarities p_i set to zero to learn a single truth direction.

In the left panel, we progressively include more statement types in the training data for $\mathbf{t}$: first affirmative, then negated, followed by logical conjunctions and disjunctions. We measure the separation of true and false activations along $\mathbf{t}$ via the AUROC. The right panel shows the separation along truth directions learned from activations $\bar{\mathbf{a}}_{ij}$ which have been projected onto the orthogonal complement of the 2D truth subspace:

$$\bar{\mathbf{a}}_{ij} = P^\perp(\mathbf{a}_{ij}), \tag{7}$$

where $P^\perp$ is the projection onto the orthogonal complement of $\text{Span}(\mathbf{t}_G, \mathbf{t}_P)$. We train all truth directions on 80% of the data, evaluating on the held-out 20% if the test and train sets are the same,

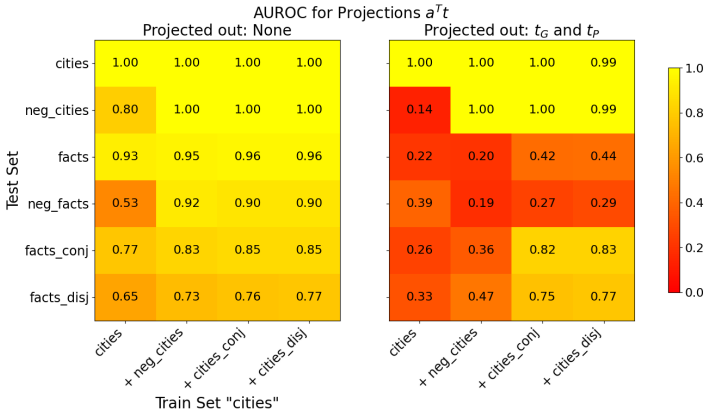


Figure 5: Generalisation accuracies of truth directions $\mathbf{t}$ before (left) and after (right) projecting out $\text{Span}(\mathbf{t}_G, \mathbf{t}_P)$ from the training activations. The x-axis is the training set and the y-axis the test set.

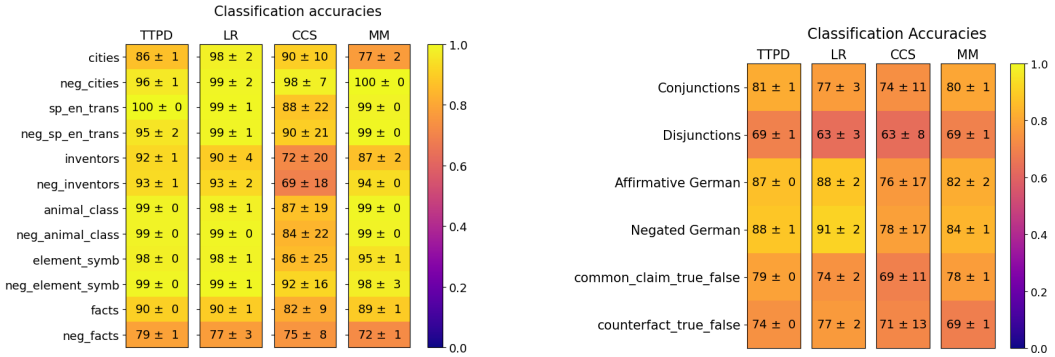
or on the full test set otherwise. The displayed AUROC values are averaged over 10 training runs with different train/test splits. We make the following observations: Left panel: (i) A truth direction $\mathbf{t}$ trained on affirmative statements about cities generalises to affirmative statements about diverse scientific facts but not to negated statements. (ii) Adding negated statements to the training set enables $\mathbf{t}$ to not only generalize to negated statements but also to achieve a better separation of logical conjunctions/disjunctions. (iii) Further adding logical conjunctions/disjunctions to the training data provides only marginal improvement in separation on those statements. Right panel: (iv) Activations from the training set `cities` remain linearly separable even after projecting out $\text{Span}(\mathbf{t}_G, \mathbf{t}_P)$. This suggests the existence of topic-specific features $\mathbf{f}_i \in \mathbb{R}^d$ correlated with truth within individual topics. This observation justifies balancing the training dataset to include an equal number of statements from each topic, as this helps disentangle $\mathbf{t}_G$ from the dataset-specific vectors $\mathbf{f}_i$. (v) After projecting out $\text{Span}(\mathbf{t}_G, \mathbf{t}_P)$, a truth direction $\mathbf{t}$ learned from affirmative and negated statements about cities *fails* to generalize to other topics. However, adding logical conjunctions to the training set restores generalization to conjunctions/disjunctions on other topics.

The last point indicates that considering logical conjunctions/disjunctions may introduce additional linear structure to the activation vectors. However, a truth direction $\mathbf{t}$ trained on both affirmative and negated statements already generalizes effectively to logical conjunctions and disjunctions, with any additional linear structure contributing only marginally to classification accuracy. Furthermore, the PCA plot shows that this additional linear structure accounts for only a minor fraction of the LLM's internal linear truth representation, as no significant third Principal Component appears.

In summary, our findings suggest that $\mathbf{t}_G$ and $\mathbf{t}_P$ represent most of the LLM's internal linear truth representation. The inclusion of logical conjunctions, disjunctions and German statements did not reveal significant additional linear structure. However, the possibility of additional linear or non-linear structures emerging with other statement types, beyond those considered, cannot be ruled out and remains an interesting topic for future research.

5 Generalisation to unseen topics, statement types and real-world lies

In this section, we evaluate the ability of multiple linear classifiers to generalize to unseen topics, unseen types of statements and real-world lies. Moreover, we introduce TTPD (Training of Truth and Polarity Direction), a new method for LLM lie detection. The training set consists of the activation vectors $\mathbf{a}_{ij}$ of an equal number of affirmative and negated statements, each associated with a binary truth label τ_{ij} and a polarity p_i, enabling the disentanglement of $\mathbf{t}_G$ from $\mathbf{t}_P$. TTPD's training process consists of four steps: From the training data, it learns (i) the general truth direction $\mathbf{t}_G$, as outlined in Section 3, and (ii) a polarity direction $\mathbf{p}$ that points from negated to affirmative statements in activation space, via Logistic Regression. (iii) The training activations are projected onto $\mathbf{t}_G$ and $\mathbf{p}$. (iv) A Logistic Regression classifier is trained on the two-dimensional projected activations.



(a) (b)

Figure 6: Generalization accuracies of TTPD, LR, CCS and MM. Mean and standard deviation computed from 20 training runs, each on a different random sample of the training data.

In step (i), we leverage the insight from the previous sections that different types of true and false statements separate well along $\mathbf{t}_G$. However, statements with different polarities need slightly different biases for accurate classification (see Figure 1). To accommodate this, we learn the polarity direction $\mathbf{p}$ in step (ii). To classify a new statement, TTPD projects its activation vector onto $\mathbf{t}_G$ and $\mathbf{p}$ and applies the trained Logistic Regression classifier in the resulting 2D space to predict the truth label.

We benchmark TTPD against three widely used approaches that represent the current state-of-the-art: (i) Logistic Regression (LR): Used by Burns et al. [2023] and Marks and Tegmark [2023] to classify statements as true or false based on internal model activations and by Li et al. [2024] to find truthful directions. (ii) Contrast Consistent Search (CCS) by Burns et al. [2023]: A method that identifies a direction satisfying logical consistency properties given contrast pairs of statements with opposite truth values. We create contrast pairs by pairing each affirmative statement with its negated counterpart, as done in Marks and Tegmark [2023]. (iii) Mass Mean (MM) probe by Marks and Tegmark [2023]: This method derives a truth direction $\mathbf{t}_{\text{MM}}$ by calculating the difference between the mean of all true statements $\boldsymbol{\mu}^+$ and the mean of all false statements $\boldsymbol{\mu}^-$, such that $\mathbf{t}_{\text{MM}} = \boldsymbol{\mu}^+ - \boldsymbol{\mu}^-$. To ensure a fair comparison, we have extended the MM probe by incorporating a learned bias term. This bias is learned by fitting a LR classifier to the one-dimensional projections $\mathbf{a}^\top \mathbf{t}_{\text{MM}}$.

5.1 Unseen topics and statement types

Figure 6a shows the generalisation accuracy of the classifiers to unseen topics. We trained the classifiers on an equal number of activations from all but one topic-specific dataset (affirmative and negated version), holding out this excluded dataset for testing. TTPD and LR generalize similarly well, achieving average accuracies of $93.9 \pm 0.2\%$ and $94.6 \pm 0.7\%$, respectively, compared to $84.8 \pm 6.4\%$ for CCS and $92.2 \pm 0.4\%$ for MM.

Next, we evaluate the classifiers' generalization to unseen statement types, training solely on activations from English affirmative and negated statements. Figure 6b displays classification accuracies for logical conjunctions, disjunctions, and German translations of affirmative and negated statements, averaged across multiple datasets. Individual dataset accuracies are presented in Figure 9 of Appendix E. TTPD outperforms LR and CCS in generalizing to logical conjunctions and disjunctions. It also exhibits impressive classification accuracies on German statements, only a few percentage points lower than their English counterparts. For the more diverse and occasionally ambiguous test sets `common_claim_true_false` and `counterfact_true_false`, which closely resemble the training data in form, TTPD and LR perform similarly well.

5.2 Real-world lies

The ultimate goal of our research extends beyond distinguishing simple true and false statements. We aim to reliably detect sophisticated lies generated by LLMs [Scheurer et al., 2024, Hubinger et al., 2024, Järviniemi and Hubinger, 2024]. In such settings we cannot trust the LLM outputs as a